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COMP4139
Machine Learning (MLE)

- Data Pre-processing & Feature Analysis

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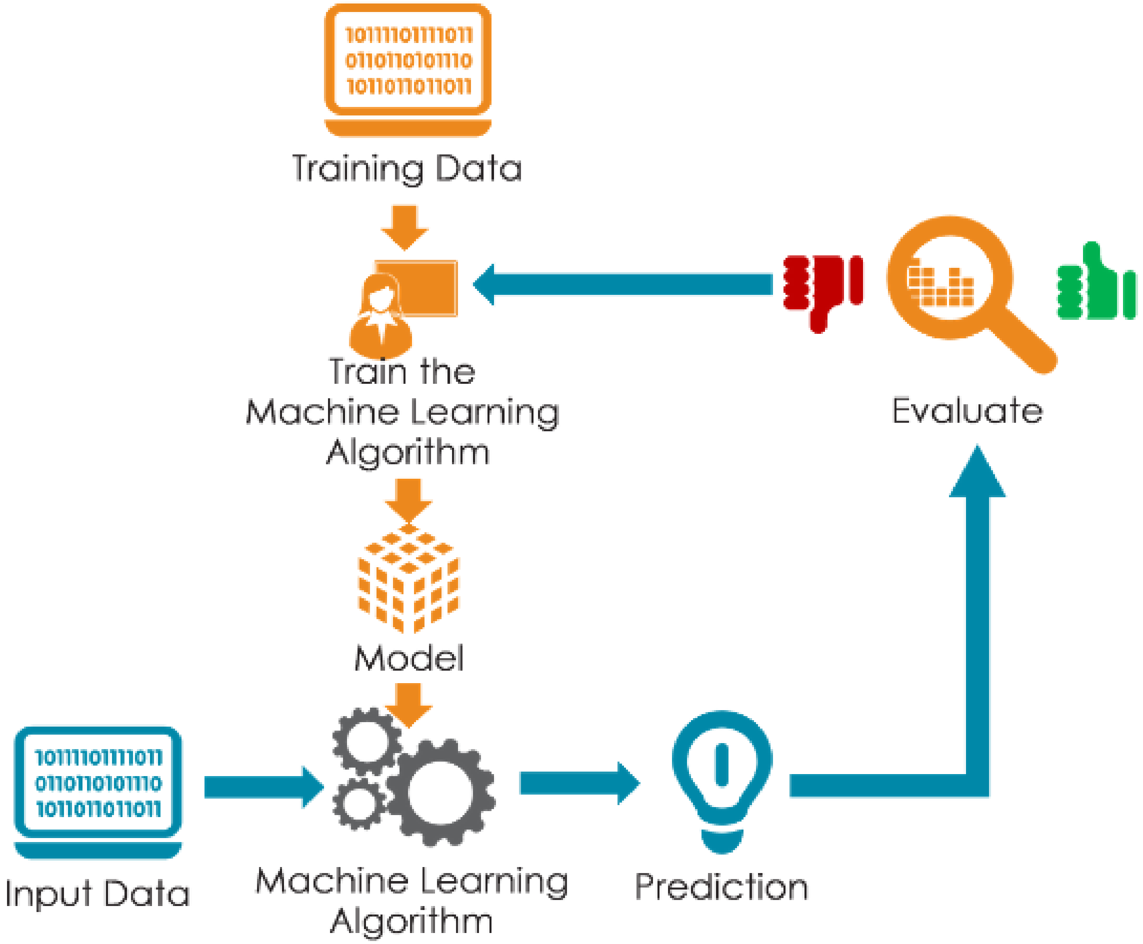
General Pipeline of Machine Learning

- Data representation + Modelling + Evaluation + Optimisation (Pedro Domingos, 2012)



Data Understanding, Representation and Pre-processing

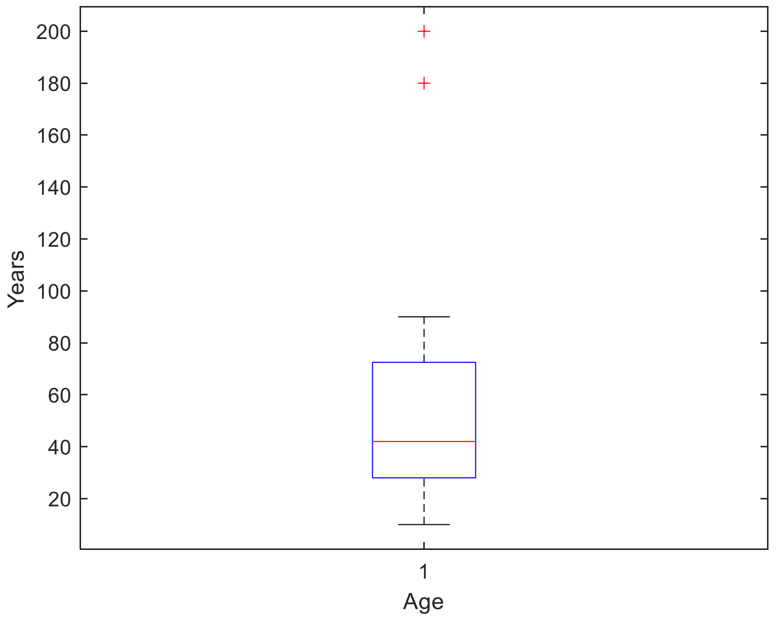
- Understand the data
  - Truly understand the underlying problem as much as possible.
  - Visualise the data (outliers, value range, etc.)

- Feature representation
  - Reliability, repeatability
  - Categorical, binary, continuous.
  - Feature value normalisation

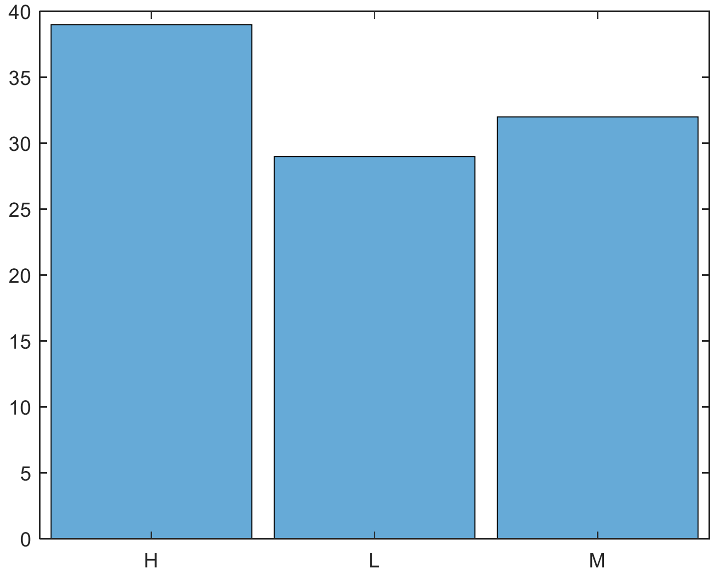
- Data pre-processing
  - Missing data, errors, etc.
  - Data imputation

Data Visualisation

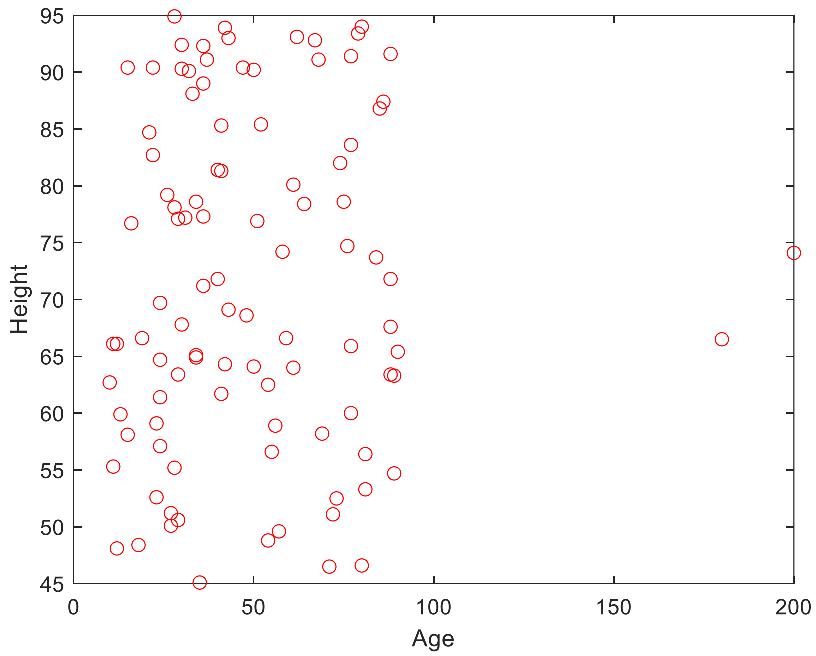
Boxplot: Good to visualise continuous variable (check outliers)



Histogram: Good to visualise categorical variables (check distribution)



Scatter plot: Good to explore relationships between two variables



Categorical to One-hot Encoding

Index	Animal
0	Dog
1	Cat
2	Sheep
3	Horse
4	Lion

Data Normalisation

- Methods:
  - Z-Normalisation (or Zero-mean normalisation)

$$X_{norm} = (X - \mu)/\sigma$$

X is the feature vector of original values, μ and σ are the mean and standard deviation of vector X.

  - Min-Max Normalisation (use 5%/95% percentile as the min and max more robust to outliers)

$$X_{norm} = (X - \min(X))/(\max(X)\text{-}\min(X))$$

Min(X) and Max(X) are the minimum and maximum values of vector X respectively

  - Vector Normalisation

$$X_{norm} = \frac{X}{||X||}$$

$||X||$ is the vector length of X

Data Normalisation

- Benefit of data normalisation:
  - ✓ They are all linear scaling methods, so won't affect the original data distribution.
  - ✓ Improves the numerical stability of the machine learning model. (e.g. using gradient descend to optimise neural network, if different features are in different value range a fixed learning rate will likely overshoot or undershoot the optima for certain features).
  - ✓ Reduce the negative impact to distance-based algorithms (e.g. KNN, SVM).

Data Imputation (deal with missing data)

- Get rid of the instances with missing features (safer when have a lot of samples)

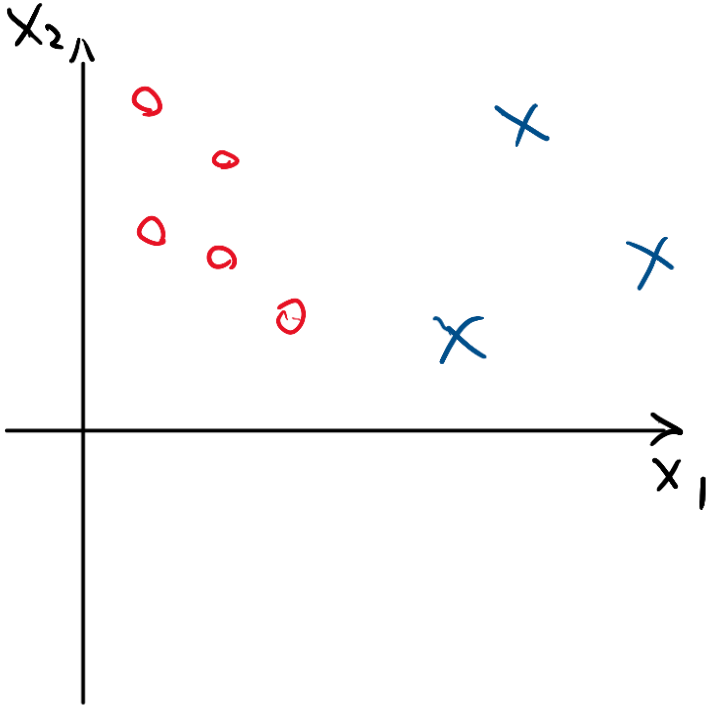
- Some algorithms will ignore the missing data, but majority of methods will be panic and output errors.

- Data Imputation methods:
  - Use mean or median values
  - Use most frequent values
  - Use k-nearest neighbour based on feature similarity
  - Use multivariate imputation by chained equations (i.e. try filling the missing data multiple times with different values and pool the final results)
  - Estimate the missing value using machine learning models based on other features.

Curse of Dimensionality

- What is curse of dimensionality in machine learning?

  - Data samples are too sparse in the feature space. (i.e. the number of instances are not large enough to densely distributed in the feature space)



Curse of Dimensionality

- How many data samples do we need to fill in the feature space ?

Curse of Dimensionality

- What shall we do to avoid curse of dimensionality?
  - Increase the number of data samples. The required sample numbers will need to increase exponentially with a linear increase of feature dimensions.
  - Reduce the number of features (more feasible): use feature selection and dimensionality reduction methods.